

Theo Usher

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EDUCATION

Columbia University, *Bachelor of Science, Mechanical Engineering* New York, NY; Sep 2020 – May 2024
GPA 4.07/4.0, Tau Beta Pi Engineering Honor Society, Mechanical Engineering Certificate of Merit, Magna Cum Laude
Electives: Robotics Studio, Managing Tech Innovation, Public Speaking, Human-Centered Design

WORK EXPERIENCE

Boeing, *Space Electronics and Mechanical Product Design Engineer* Los Angeles, CA; Aug 2024 – Present

- Serve as Mechanical Responsible Engineering Authority for Spacecraft Control Electronics
- Drive flight hardware design from concept through prototyping, testing, and production
- Develop mechanical designs that balance structural, thermal, and manufacturing constraints
- Design printed circuit boards with over 25 layers and thousands of components
- Release production-ready drawing packages with GD&T callouts and manufacturing specifications
- Diagnose and resolve dozens of flight hardware discrepancies through root cause analysis and corrective actions
- Lead production support operations, coordinating engineers and dispositioning millions in flight hardware

Airobotics Drones, *Design Engineering Intern* Tel Aviv, Israel; May 2023 – Jul 2023

- Designed mechanical systems for the Iron Drone, an autonomous counter-drone
- Analyzed, tested, and iterated on the drone's net launcher for range and reliable interception
- Redesigned the drone ground station for improved functionality, reliability, and environmental protection
- Prototyped and optimized the drone ground station, cutting assembly time from 2 weeks to 2 days
- Engineering contributions used in Iron Drone systems deployed globally and exhibited at the U.S. DOT

Terabase Energy, *Mechanical Engineering Intern* Davis, CA; May 2022 – Aug 2022

- Designed and field-tested robotic vehicles to transport and install solar panels in utility-scale projects
- Designed a solar panel installation mechanism that more than doubled installation range
- Built a panel stabilization system that replaced high-cost linear actuators with low-cost, durable mechanisms
- Engineering contributions deployed in autonomous factories across multiple utility-scale solar installations

ENGINEERING PROJECTS & LEADERSHIP

Columbia Space Initiative, *Rockets Mission Lead* Sep 2020 – Jun 2024

- Led a 50-person team to research, design, and build a hybrid rocket to reach 30,000 ft with a scientific payload
- Iterated rapidly, achieving a rare speed of designing, building, and launching a new rocket every year
- Drove cross-subsystem integration across hardware, software, and electromechanical systems
- Oversaw manufacturing across multiple processes, including CNC machining, 3D printing, and composites layup
- Designed, manufactured, and tested parachute deployment mechanisms for rocket recovery
- Created and ran test procedures, including for rocket static fires and recovery testing
- Achieved Rockets' first successful flight and recovery, improved engine performance by 10x
- Won President's Choice Award at Spaceport America Cup 2023 and Best Podium Presentation at FAR-OUT 2024
- Cultivated a team that, a year later, became the first US student-led group to launch a liquid-oxygen hybrid rocket

Bipedal Walking Robot Jan 2022 – May 2022

- Designed, programmed, and optimized a bipedal robot from scratch using a Raspberry Pi
- Achieved the first and fastest walking robot in a class where half the students could not finish

Automated Robotic Linkage – Machine Design Sep 2023 – Dec 2023

- Designed, built, and programmed a complex linkage mechanism to rapidly press arcade buttons
- Created a unique, cable-driven mechanism unlike any that our professor had seen, greatly reducing inertia

SKILLS

Technical Skills: CAD (SolidWorks, Creo, Fusion), PCB Design, 3D Printing, CNC Machining, Sheet Metal Forming, FEA, GD&T, CAM, DFM, DFA, PDM, MATLAB, Python, C++, Root Cause Analysis, Prototyping, Hand Tools, Soldering

Non-Technical Skills: Leadership, Stakeholder Management, Cross-Functional Collaboration, Communication